

Subject Description Form

Subject Code	ITC2002T
Subject Title	Textile Studies I
Credit Value	3 credits
Level	2
Pre-requisite/ <Co-requisite> / (Exclusion)	Nil
Objectives	The subject aims for providing a fundamental theoretical, professional, and technological knowledge of polymer, fiber and yarn in textile.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Explain and apply the structure-property relationship of polymers used in textile fibers and yarns. b. Compare the performance of different textile materials. c. Describe the origins, production, properties and end-uses of common textile fibers and relate the different characteristics of the various fibers to their applications. d. Identify the general types of fiber and yarn and their properties. e. Compare the principal methods of yarn production and relate them to the general types of yarn and their properties.
Subject Synopsis/ Indicative Syllabus	(I) Polymer: Basic terminology in polymer science, structure of polymer, typical thermal transition of polymer, physical properties of polymer (II) Fiber: The origin, production, properties, and end-uses of different natural and man-made fibers, fiber quality, identification (III) Yarn: Classification, structure, properties and uses of spun yarns, continuous filament yarn, texturized yarn, fancy yarn, principles underlying the processes used in the conversion of fibers to yarn, staple yarn spinning systems (outlines), yarn count and yarn twist (IV) Testing: Fiber testing (identification of fibers, determination of fiber length, finesses, maturity and tensile strength); yarn testing (assessment of linear density, yarn twist, strength and irregularity)
Teaching/Learning Methodology	By lecture, tutorial, and self-study.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c	d	e	
	1. Coursework	50%	✓	✓	✓	✓	✓	
	2. Examination	50%	✓	✓	✓	✓	✓	
	Total	100%						
Student Study Effort Expected	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:							
	This subject provides fundamental theory and some basic laboratory trainings to students who would like to understand textiles. Therefore, it is necessary to carry out coursework (such as laboratory reports, group projects, lecture quiz), to assess the student’s learning outcomes at different stages and aspects of the subject, as well as to performance final examination to assess the overall understanding of the theory.							
	Class contact:							
	▪ Lecture						26 Hrs.	
	▪ Lab						18 Hrs.	
	Other student study effort:							
Reading List and References	▪ Self-study						62 Hrs.	
	Total student study effort						106 Hrs.	
	<u>Books</u>							
	Hoboken, N.J. (2006). <i>Introduction to Physical Polymer Science</i> . 4 th Ed. Wiley.							
	Bower, D.I. (2002). <i>An Introduction to Polymer Physics</i> . Cambridge University Press.							
	Chan, K., Cheng, K.P., and Lau, M.P. (1993). <i>Understanding Textiles, Volume 1: Fibre Technology</i> . The Hong Kong Polytechnic University.							
Collier, B.J. and Tortora, P.G. (2001). <i>Understanding Textiles</i> . NJ: Prentice Hall.								
Hudson, P.B., Clapp, A.C. and Kness, D. (1993). <i>Joseph’s Introductory Textile Science</i> . Fort Worth: Harcourt Brace Jovanovich College Publishers.								
Gong, R.H. (2002). <i>Fancy Yarns: Their Manufacture and Application</i> . CRC Press/Woodhead Pub.								
Klein, W. (1993). <i>New Spinning System</i> . Manchester: Textile Institute.								
Laurence, C.A. (2003). <i>Fundamentals of Spun Yarn Technology</i> . FL: CRC Press.								
Gong, R.H. and Wright, R.M. (2002). <i>Fancy Yarns: Their Manufacture and Application</i> . CRC Press/Woodhead Pub.								